



TAXI TRAJECTORY PREDICTION

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Engineering Production-Ready ML/AI Systems

MOTIVATION

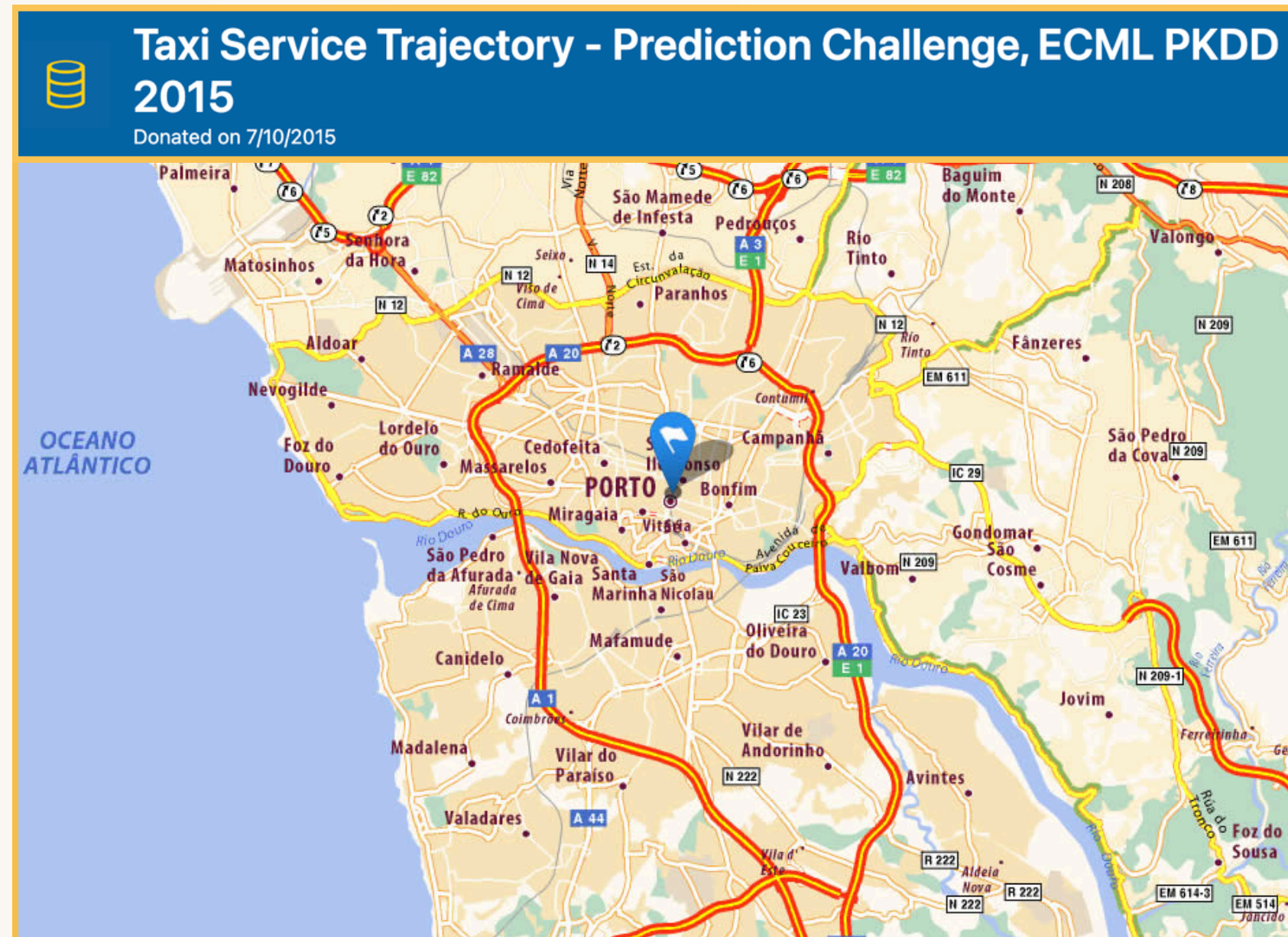
Urban mobility systems (ride-hailing, dispatching, fleet optimization) need to anticipate where vehicles are going next

Raw GPS trajectories are noisy, large-scale, and not ML-ready

We want actionable predictions for:

- Better dispatching
- Reduced idle time
- Smarter routing & congestion avoidance

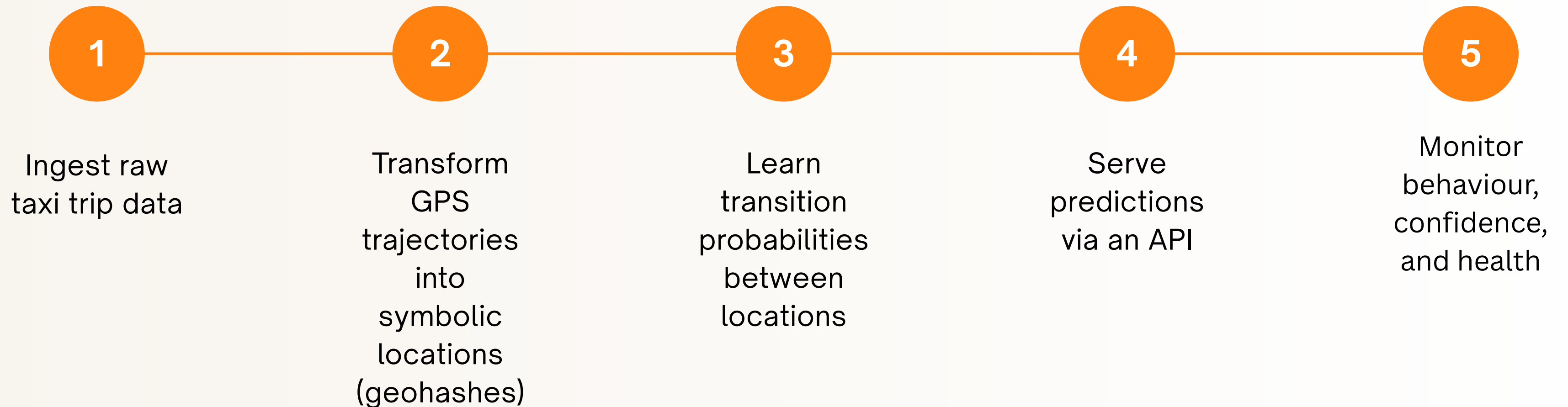
DATASET



Given a partial trajectory of a taxi trip, can we predict the next spatial location it will move to?

IDEA

Given a partial trajectory of a taxi trip, can we predict the next spatial location it will move to?



BUILD >>> ML Pipeline

Distributed Data
(Pre)processing

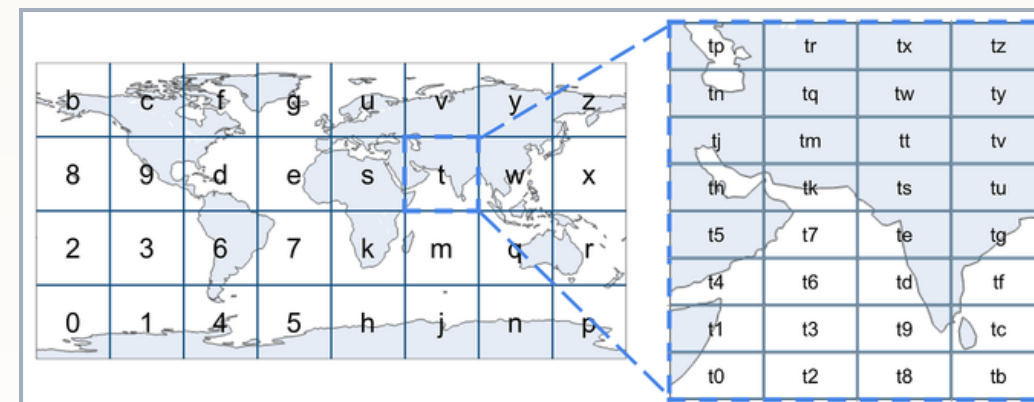


Batch Training
Orchestration



GPS polyline
trajectories

+ hour
+ day of the week
+ call type

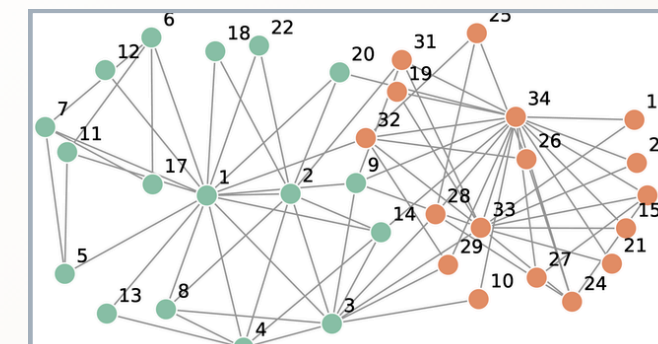


GeoHash (prev, curr → next)
transitions

+ temporal encodings
(cyclical and semantic)
+ call type



Empirical movement
graph



Logistic Regression
Top-K next locations

Prediction &
Artefacts

BUILD >>> ML Pipeline

Batch Training
Orchestration



*everything is containerized in Docker and managed with Docker compose

BUILD >>> Model Serving



GET /health

POST
/predict_next_node

GET
/graph_node_info
/{node}

GET /metrics

BUILD >>> Monitoring & Telemetry

Metrics + Alerts  Prometheus

taxi-api-alerts /etc/prometheus/prometheus-alerts.yml last run 20.627s ago took 4ms every 1m

🔔 TaxiApiDown

20.728s ago 3ms OK ^

```
up{job="taxi_api"} == 0
```

for: 30s

severity="critical"

summary

Taxi API is DOWN

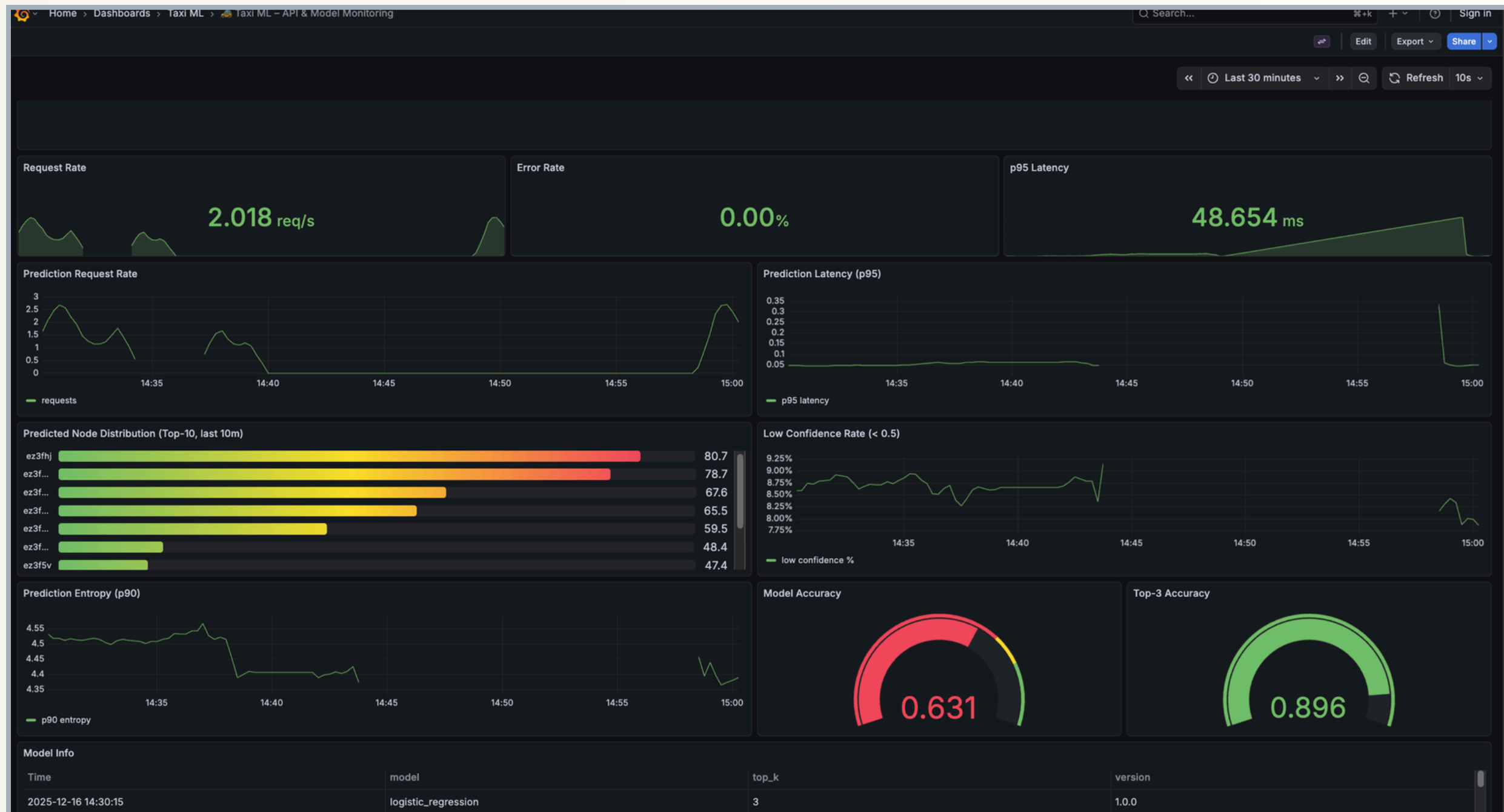
🔔 HighErrorRate

20.725s ago 0ms OK v

🔔 HighLatency

20.725s ago 0ms OK v

BUILD >>> Monitoring & Telemetry



Dashboard





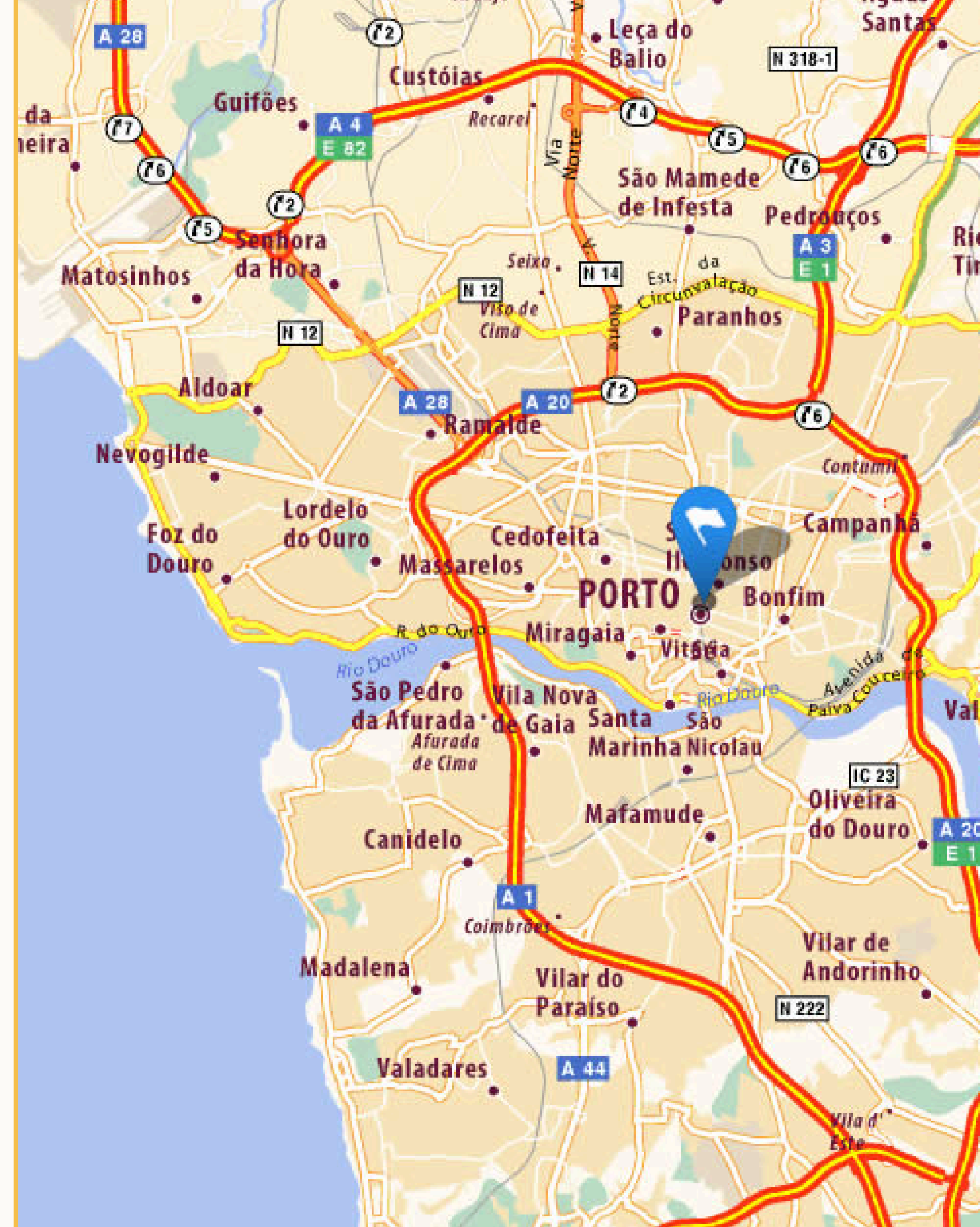
DEMO



100%

This project demonstrated:

- Full MLOps lifecycle
- Separation of concerns:
 - Data
 - Training
 - Serving
 - Monitoring
 - Testing
- Production-oriented decisions





THE END

Thank you! Questions?